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Unit 3 - Building a Smart Slide for Playground Safety

PDF Documents

PDF Document 1: 1738217442733-Chapter 3 - Unit 3 - Building a Smart Slide for Playground Safety.pdf

This PDF document is included in the unit materials.

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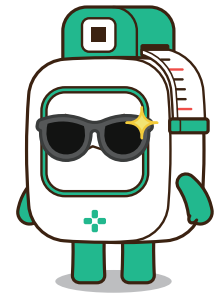


UNIT 3 BUILDING A SMART SLIDE FOR PLAYGROUND SAFETY



In this project, you will create a Smart Slide that helps keep everyone safe while they play. The Smart Slide will use a distance sensor to detect if someone is near the bottom of the slide. When the slide is clear, a green light will turn on, signaling that it is safe to slide. However, if someone is too close, a red light will turn on, warning other users to wait until the area is clear.

Let's explore how you can build this safety system using the MODI Master Kit!



HOW THE SMART SLIDE WORKS

The Smart Slide relies on two main components:

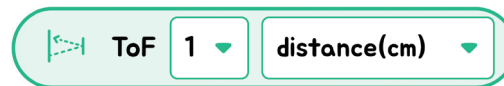
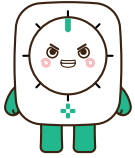
ToF Sensor (Time of Flight): Measures the distance between the slide and any object or person in front of it, detecting if someone is too close to the exit.

LED Light: Changes color based on the sensor's reading. If someone is too close, it turns red to signal "wait." If the area is clear, it turns green, indicating it's safe to slide.

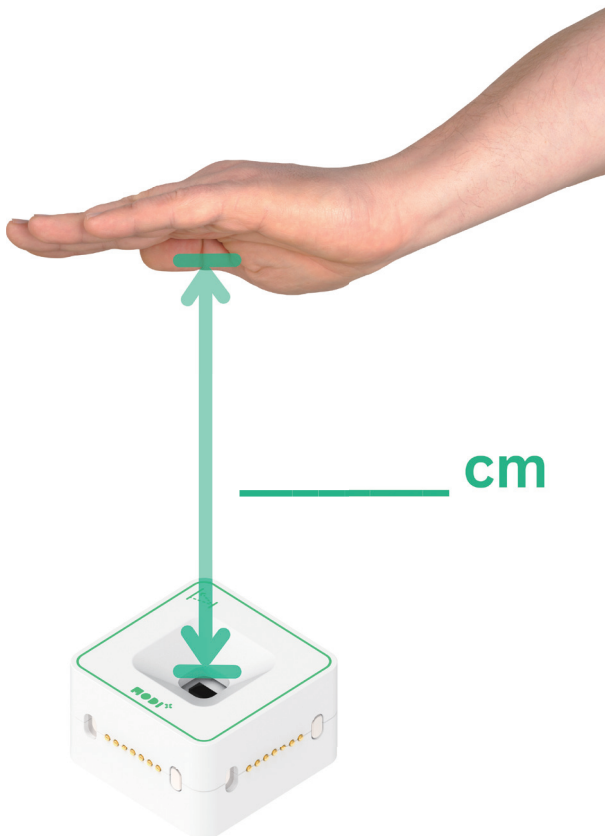
will turn green, indicating it's safe to slide.



Do you remember what ToF can do?



✓ distance(cm)
distance(inch)



Main Features of ToF

- Distance Measurement
It measures the distance to an object by timing how long it takes for a light signal to travel to the object and back.
- Real-time Processing
Capable of measuring distances in real-time, which is essential for dynamic environments and applications like robotics and autonomous vehicles.

What is the distance between the ToF module and your hand now?



Material Needed



Network

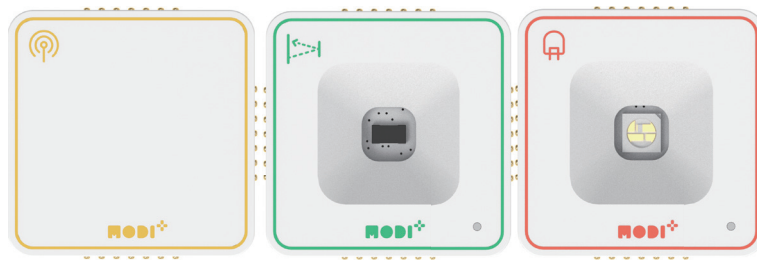


TOF



LED

When we combine all the modules, a creation has to look like this.



Code Story

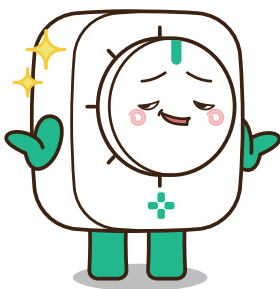
When ToF's distance is less than 15cm

T ↓

Set LED as **red**

F ↓

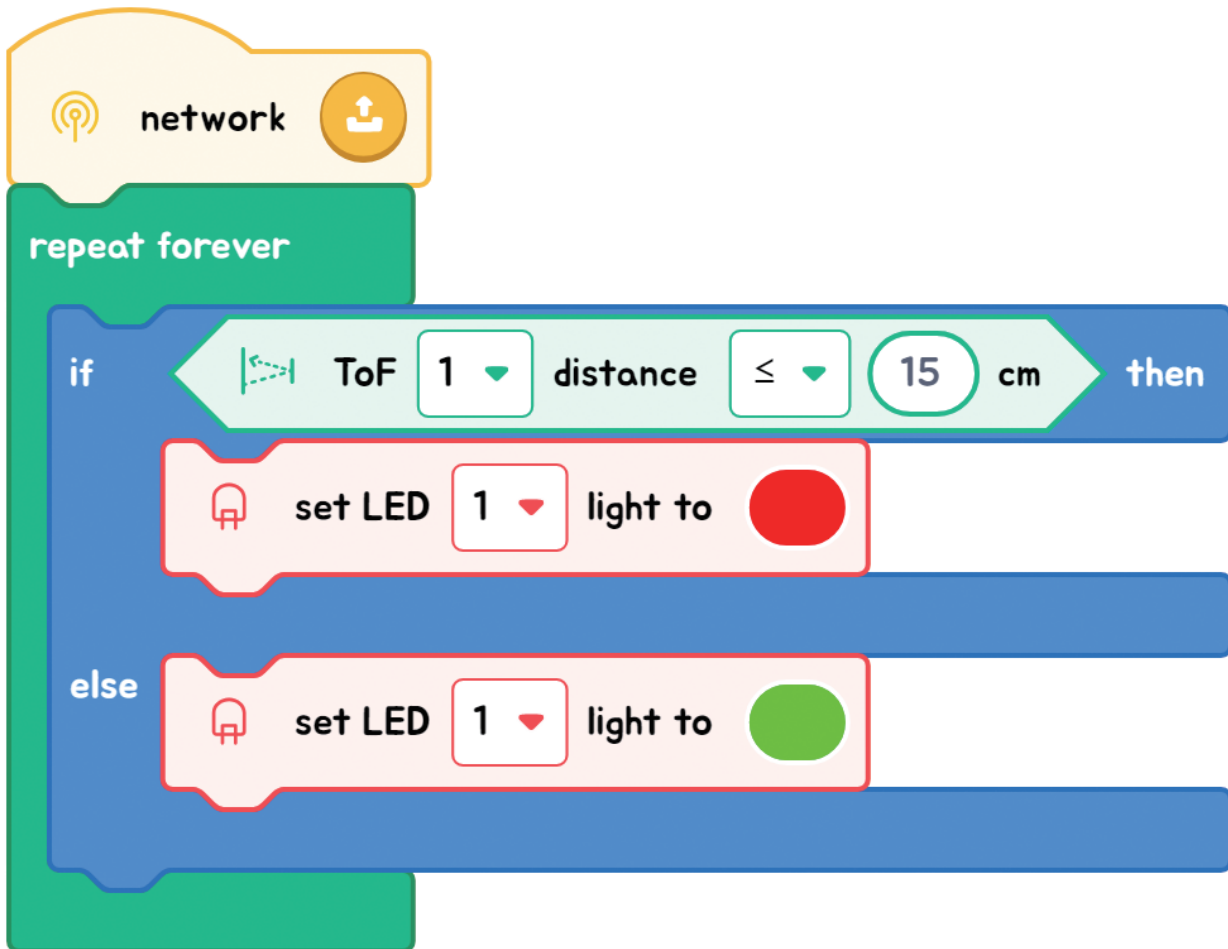
Set LED as **green**



T: True
F: Fales



Code Quest



Checkpoint

Let's add code speaker when there a person neary ToF to make a alarm sound..

